



Mg and Al mixed oxides and the synthesis of *n*-butanol from ethanol

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ABSTRACT

n-Butanol is used in a wide range of applications, especially for the production of paint, solvents and plasticizers. Recently, some companies have proposed the use of *n*-butanol as a fuel, which can be employed pure or blended with gasoline or diesel. At present, there is great interest in developing a one-step process that generates *n*-butanol from ethanol. Some works showed that Mg–Al mixed oxides are very promising catalysts for this matter. Analyzing the physical–chemical properties of these catalysts, this work aims at further studying this reaction. The Mg–Al mixed oxides were obtained by thermal decomposition of hydrotalcites employing two Mg/Al different ratios. Magnesium oxide (MgO) and alumina (Al₂O₃) were also prepared following the same synthetic route used for the Mg–Al mixed oxides. The synthesized samples were characterized by the following techniques: XRD, NMR, BET and XRF. The acid and basic properties of the samples were also analyzed employing probe molecules. The catalytic tests were performed using a fixed bed reactor at atmospheric pressure. It was verified that the Mg and Al mixed oxides are able to promote the synthesis of C4 compounds from ethanol. Adjacent acid and medium basic sites are needed in order to generate the intermediate compounds. These pairs of acid and medium basic sites are obtained by the insertion of Al in the MgO lattice or in consequence of the presence of Mg in the γ -Al₂O₃ lattice. It was observed that the higher the concentration of Mg is, the higher the hydrogenation capacity of the catalyst, since a greater selectivity to *n*-butanol was observed. Strong basic sites and a specific superficial atomic arrangement seem not to be essential for the synthesis of C4 from ethanol.

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1. Introduction

Nowadays, many companies are showing a growing interest in investments focused on sustainable business from an economical, environmental and social standpoint. In relation to the Chemical Industry, special emphasis has been given to the use of renewable raw material (biomass) for the generation of final or intermediate products. According to this new picture, ethanol stands out as an important industrial raw material. Brazil and the US are in a position of great advantage due to the sheer volume of ethanol produced in these countries.

Recently, some companies have proposed the use of *n*-butanol as fuel. This new application is associated with some properties among which the energetic value of *n*-butanol (29.2 MJ/L) stands out since it is higher than the one of ethanol (19.6 MJ/L). This compound can also be used in a wide range of applications, especially for the production of paint, solvents and plasticizers [1–3].

The Oxo process is the most employed one for the *n*-butanol synthesis. It involves several reaction steps, some of which use

homogeneous catalysts and non-renewable resources such as propylene and CO [3].

At present, there is great interest in developing a one-step process that generates *n*-butanol from ethanol. This synthesis would be carried out using only one reactor where the dehydrogenation of ethanol would occur producing acetaldehyde. After that, two molecules of this aldehyde would be condensed generating acetal-dol (3-hydroxybutyraldehyde), which would then be dehydrated and hydrogenated producing *n*-butanol [4].

So far, the two most promising catalytic systems, which have been proposed regarding this synthesis are the following: the hydroxyapatites and the mixed oxides obtained from hydrotalcites composed of Mg and Al [4–10]. The first system shows high selectivity to *n*-butanol at low conversion levels [5,6]. However, at high conversions, hydroxyapatites generate compounds with high molecular weight, which provoke the deactivation of these catalysts. On the other hand, the mixed oxides are less active than the first ones, and as a result, they generate less heavy compounds. Thus, the deactivation rate is much lower.

León et al. [9] employing Mg–Al mixed oxides suggested that the catalytic systems that show high density of strong basic sites are the most selective to *n*-butanol. They also observed a very high selectivity to ethylene, which is associated with the presence of acid

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sites on the surface of these mixed oxides. Marcu et al. [10] using Cu–Mg–Al also verified that the selectivity to *n*-butanol increases when the density of strong basic sites increases as well.

Di Cosimo et al. [4], also employing Mg–Al mixed oxides, suggested that acetaldehyde condensation occurs between adjacent adsorbed acetaldehyde species. They also proposed that the formation of *n*-butanol requires not only basic sites but also acid sites. The latter is related to acetaldehyde adsorption and the former to hydrogen abstraction. The basic sites generate carbanion species, which react with adsorbed acetaldehyde producing acetaldol. This compound is promptly dehydrated and hydrogenated generating *n*-butanol.

According to some authors [4,5,11] not only the acid–basic surface properties are relevant for the aldol condensation but also a specific surface atomic arrangement composed of two acid sites and one basic site is needed in order to adsorb the acetaldehyde molecules and abstract hydrogen from one of them.

As it can be observed, these authors' suggestions are not fully in agreement. Therefore, analyzing the physical–chemical properties of the Mg–Al mixed oxides, this work aims at further studying the one-step process that generates *n*-butanol from ethanol.

2. Experimental

2.1. Catalysts and oxides preparation

The Mg–Al hydrotalcite precursors of the Mg–Al mixed oxides were prepared by the co-precipitation method. Two aqueous solutions, one comprising K_2CO_3 and KOH, and the other $Mg(NO_3)_2$ and $Al(NO_3)_3$, were prepared. Both solutions were simultaneously added dropwise to deionized water kept at 70 °C under constant stirring. The suspension obtained was filtered under vacuum, washed with deionized water and dried at 120 °C for 18 h. The solid obtained was then calcined at 400 °C (10 °C min^{−1}) for 4 h and at 500 °C for 1 h. Magnesium oxide (MgO) and alumina (Al_2O_3) were also prepared following the same synthetic route used for the Mg–Al mixed oxides.

2.2. Characterization

The measurements of the specific surface area of the mixed oxides were carried out using an ASAP 2010 Micromeritics instrument. The samples were treated at 100 °C for 24 h. After that, they were submitted to a treatment in situ under vacuum at 150 °C for 2 h. The N_2 adsorption was carried out at −196 °C.

The Mg/Al atomic ratios of the synthesized catalysts were determined by X-ray fluorescence (XRF) using a Rigaku RIX 3100 spectrometer.

The X-ray diffraction (XRD) was performed using a Philips diffractometer, model D5000 Siemens, operating with radiation Cu $K\alpha$ (40 kV, 30 mA) and a Ni filter. The angular range varied from 5° to 80°, with increments of 0.05° and 5 s per step. Diffractograms were analyzed by the Rietveld method with fundamental parameters employing the TOPAS program (Bruker, version 4.2).

The acidity of the samples was determined by pyridine adsorption experiments followed by FTIR spectroscopy. The spectra were recorded from self-supporting wafers using a Nicolet Magna spectrophotometer. The samples were pretreated at 500 °C for 1 h under vacuum, submitted to a stream of air for 30 min and finally exposed to high vacuum for 30 min (10^{−7} Torr). Pyridine was adsorbed at room temperature, at 10 Torr for 1 h. Spectra were collected after desorption at room temperature for 1 h under high vacuum. The spectra were normalized to the mass of the wafers. The quantitative analysis of the Lewis acid sites was done considering the vibrations around 1450 cm^{−1}.

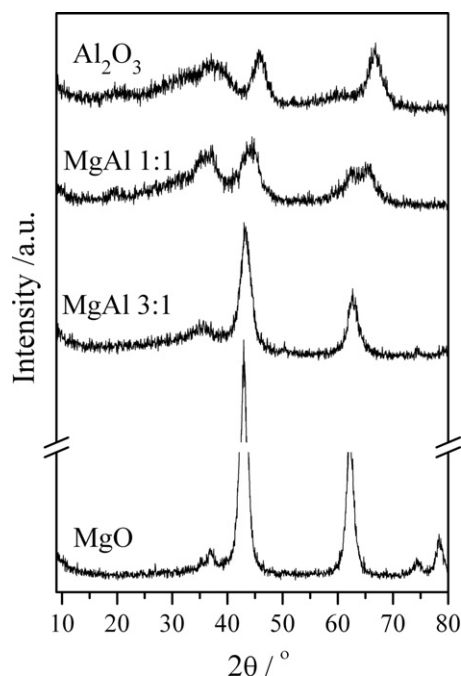


Fig. 1. XRD patterns of the synthesized catalysts.

The basic sites were analyzed by temperature-programmed desorption of CO_2 (TPD of CO_2) using a micro reactor coupled to a Micromeritics Autochem II. The samples were pretreated under 8% O_2 /He (vol.%) flow (30 mL min^{−1}) at 500 °C for 1 h. The CO_2 adsorption was conducted at room temperature for 1 h (25 mL min^{−1}). The TPD of CO_2 was carried out heating the sample at 10 °C min^{−1} up to 773 K, under He flow (50 mL min^{−1}). The signal was monitored using a TCD detector. The desorption profiles were deconvoluted using Gaussian curves in order to quantify the weak, medium and strong basic sites. The weak strength sites are related to a curve which shows a maximum at temperature lower than 170 °C, the ones between 170 and 270 °C are assigned to medium strength basic sites, and finally, the ones above 270 °C, strong basic sites.

MAS (magic-angle spinning) ^{27}Al NMR (nuclear magnetic resonance) spectra were obtained on a Varian Infinity-Plus-400 spectrometer at room temperature, at 103.9 MHz with zirconia rotors (4 mm diameter) and a sample spinning rate of 10,000 Hz. The length of the rf pulses was 1.0 μ s. All measurements were carried out at room temperature employing $AlCl_3 \cdot 6H_2O$ as a standard reference (0 ppm).

2.3. Catalytic testes

The catalytic tests were performed using a conventional system with a fixed bed reactor at atmospheric pressure and monitored by on-line gas chromatography (CG 6850). A 12% ethanol/ N_2 (vol.%) mixture was fed at 40 mL min^{−1} on a catalyst mass of 300 mg. A binary physical mixture comprised of Al_2O_3 and ZrO_2 (with 300 mg of each catalyst) was also employed. Ethanol vapors were generated by passing nitrogen through a saturator at 30.5 °C. The catalysts were pretreated at 500 °C for 1 h under synthetic air. The deactivation rate during the catalytic tests (12 h) was very low. Thus, the values of conversion and selectivities were calculated considering the average values of the last 8 h.

Table 1

The Mg/Al atomic ratio obtained by X-ray fluorescence ($\text{Mg/Al}_{\text{XRF}}$), the concentration of crystalline species (%p/p) and the Mg/Al atomic ratio ($\text{Mg/Al}_{\text{XRD}}$) obtained by XRD (Rietveld method).

Samples	$(\text{Mg/Al})_{\text{XRF}}$	XRD		
		Crystalline species	%p/p	$(\text{Mg/Al})_{\text{XRD}}$
MgAl 1:1	1.2	MgO	7.2	0.2
		$\gamma\text{-Al}_2\text{O}_3$	70.5	
		MgAl_2O_4	22.3	
MgAl 3:1	3.0	MgO	90.6	12.3
		$\gamma\text{-Al}_2\text{O}_3$	9.4	
Al_2O_3	–	Al_2O_3	100	–
MgO	–	$\gamma\text{-MgO}$	100	–

3. Results and discussion

Fig. 1 and Table 1 display the results related to the X-ray diffraction of the synthesized samples. The sample containing the highest percentage of Mg shows a predominance of MgO (cubic). In the case of the catalyst with the lowest amount of this element (MgAl 1:1), the predominant phase is $\gamma\text{-Al}_2\text{O}_3$. This sample also shows the presence of MgAl_2O_4 and MgO. In the case of Al_2O_3 and MgO samples, $\gamma\text{-Al}_2\text{O}_3$ and MgO (cubic) were observed, respectively. Both mixed oxides showed XRD patterns assigned to hydrotalcite structures before the calcination step.

The concentration of crystalline species shown in Table 1 refers only to the crystalline nature of these materials. It was not possible to determine the composition of the structures mentioned above due to the following reasons: the proximity of the atomic numbers of Mg and Al, the nanocrystalline nature of the catalysts, and the presence of several overlapping reflections.

Table 1 also shows two Mg/Al atomic ratios, one obtained by chemical analyses ($\text{Mg/Al}_{\text{XRF}}$) and the other through XRD results ($\text{Mg/Al}_{\text{XRD}}$). It can be verified that in the case of the MgAl 3:1 sample, the Mg/Al ratio inferred by XRD is much higher than the one obtained by the chemical analyses. Moreover, the Mg/Al ratio obtained by XRD in the case of the MgAl 1:1 catalyst, is lower than the one from chemical analysis. Taking these Mg/Al values into account, it can be suggested that in the case of the MgAl 3:1 sample, Al is in the lattice of MgO. And for the MgAl 1:1 sample, Mg would be in the $\gamma\text{-Al}_2\text{O}_3$ lattice. In other words, 90.6% of the MgO inferred by XRD, in the case of MgAl 3:1 sample, refers to the MgO structure containing Al, and 70% of $\gamma\text{-Al}_2\text{O}_3$ observed in the case of the MgAl 1:1 sample, is associated with the presence of Mg in this structure. The spinel structure is shown by MgAl_2O_4 and Al_2O_3 and both are composed of Mg and Al. Thus, to simplify the discussion when MgAl 1:1 is studied, MgAl_2O_4 is considered together with Al_2O_3 .

Table 2 and Fig. 2 depict the NMR analysis (^{27}Al signal) of the synthesized samples. It can be noticed that only Al^{3+} cations octahedrally (Al_o) coordinated to oxygen, are observed in the case of the MgAl 3:1 sample. Taking into account that the Mg coordination in the MgO crystal is octahedral [12], this result suggests that the Al in this sample would be almost entirely in the MgO lattice, thus confirming the XRD results. On the other hand, for the MgAl 1:1 and Al_2O_3 samples, two signals were observed: one related to Al^{3+} cations tetrahedrally (Al_t) coordinated to oxygen, and another,

Table 2

^{27}Al NMR analysis of mixed oxides and Al_2O_3 samples.

Samples	Al_o	Al_t	$\text{Al}_\text{o}/\text{Al}_\text{t}$
MgAl 1:1	69.2	30.8	2.2
MgAl 3:1	100.0	–	–
Al_2O_3	81.4	18.6	4.4

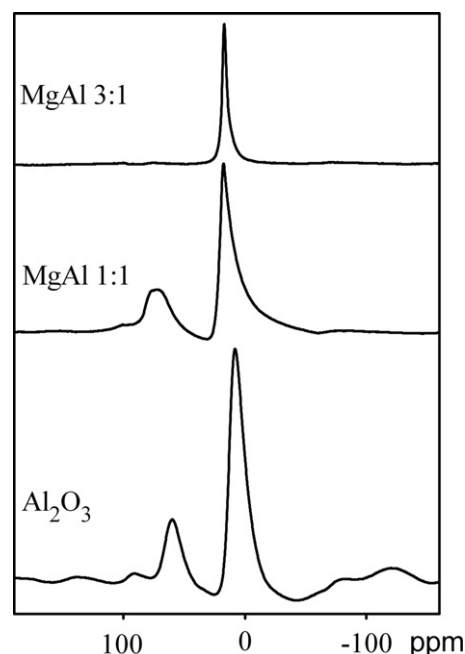


Fig. 2. ^{27}Al NMR analyses of mixed oxides and Al_2O_3 samples.

assigned to Al^{3+} cations octahedrally (Al_o) coordinated to oxygen. The $\text{Al}_\text{o}/\text{Al}_\text{t}$ ratio of MgAl 1:1 is smaller than the one of Al_2O_3 . This result suggests that Mg is in octahedral position in the $\gamma\text{-Al}_2\text{O}_3$ lattice.

Employing NMR and studying alcohol elimination reactions, Díez et al. [13] also analyzed these mixed oxides and observed the insertion of Al in the MgO structure. However, they did not obtain a structure that showed only Al^{3+} cations octahedrally (Al_o) coordinated to oxygen.

Fig. 3 depicts the TPD of CO_2 of the mixed oxides (MgAl 1:1 and MgAl 3:1), MgO and Al_2O_3 . Table 3 shows the quantification of the basic sites. As it can be observed in Fig. 3, all the samples analyzed depicted two kinds of weak basic sites (the best fit of the experimental data). As expected, Al_2O_3 only exhibits weak sites [14], whereas weak, medium and strong sites were observed in the case of MgO. When Al is added into the MgO crystal (MgAl 3:1), the density of the basic sites decreases. The strong basic sites disappear and the percentage of medium basic sites increases. On the other hand, when Mg is added to $\gamma\text{-Al}_2\text{O}_3$ (MgAl 1:1), it is possible to observe the appearance of medium basic sites.

Table 3 displays the density of the Lewis acid sites of the prepared catalysts. The MgO sample does not show acid sites. Acid sites are generated when Al is added into MgO lattice (MgAl 3:1). On the other hand, the density of acid sites decreases when Mg is added to Al_2O_3 (MgAl 1:1). Both mixed oxides show very low density of Brönsted acid sites.

These results clearly show that the insertion of Al into the MgO crystal and also Mg into the $\gamma\text{-Al}_2\text{O}_3$ lattice change the acid and basic properties of these oxides. It is interesting to verify that the values of the surface area (Table 3) almost do not change when comparing MgAl 3:1 with MgO and MgAl 1:1 with Al_2O_3 .

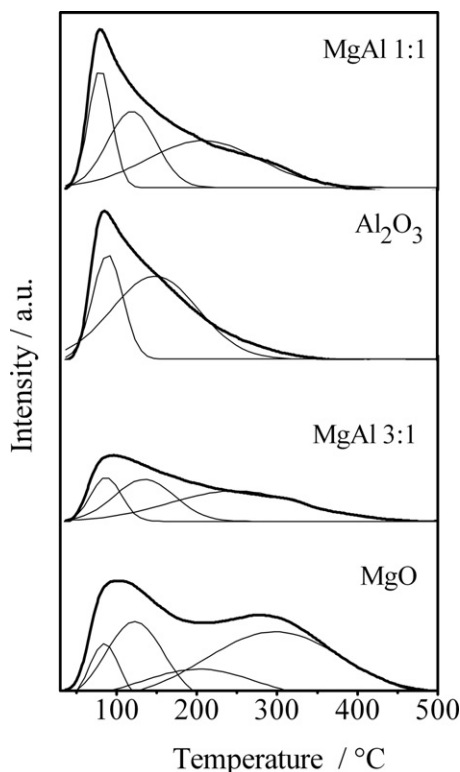
Fig. 4 shows the catalytic test behavior of the mixed oxides as well as $\text{MgO} + \text{Al}_2\text{O}_3$ physical mixture, Al_2O_3 , and MgO. Both catalysts with different Mg/Al ratios are able to condense ethanol producing *n*-butanol and butyraldehyde (C4 compounds). High concentration levels of Mg (MgAl 3:1) lead to a slightly lower activity (lower conversion), and, as a result, this catalytic system shows an increase in selectivity to acetaldehyde and a small decrease of the selectivity toward the condensation products (C4 compounds).

Table 3

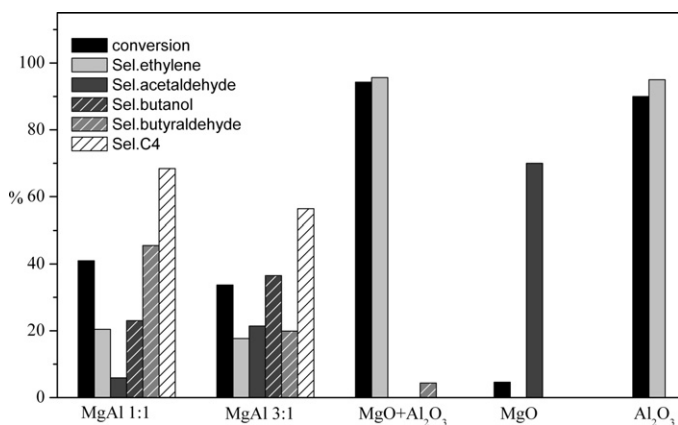
Specific surface area (S), density of weak basic sites (W_b), density of medium basic sites (M_b), strong basic sites (S_b), density of basic sites (T_b) and density of Lewis acid sites (T_a).

Samples	S ($\text{m}^2 \text{g}^{-1}$)	Density of basic sites ($\mu\text{mol g}^{-1}$)				Density of Lewis acid sites (a.u.)
		W_b	M_b	S_b	T_b	
MgAl 1:1	295	21.0 (44%)	27.0 (56%)	–	48.0	0.2
MgAl 3:1	252	13.0 (42%)	18.0 (58%)	–	31.0	0.2
Al_2O_3	309	43.3 (100%)	–	–	43.3	0.3
MgO	242	19.4 (31%)	14.5 (23%)	29.0 (46%)	62.9	–

Parentheses are the percentage of contribution of each kind of basic site.

**Fig. 3.** TPD spectra of CO_2 of the synthesized catalysts.

The higher the concentration of Mg is, the higher the hydrogenation capacity of the catalyst, since a greater selectivity to *n*-butanol was observed (Fig. 4). Both catalysts depict low selectivity to ethylene.

**Fig. 4.** Conversion of ethanol and selectivities for ethylene, butyraldehyde, *n*-butanol, acetaldehyde and C4 compounds. Experimental conditions: 350 °C, 12% ethanol/ N_2 (vol.%) at 40 mL min^{-1} , catalyst mass of 300 mg.

The synthesis of C4 compounds from ethanol is well described in the literature [4,15–17]. The first step is the generation of acetaldehyde, which occurs on pairs of acid and strong basic sites. The former is related to ethanol adsorption and the latter to hydrogen abstraction. The second step is the generation of carbanion species, which also occurs on adjacent acid and basic sites. Acid sites adsorb the acetaldehyde molecules and the basic sites abstract the proton from this aldehyde, generating carbanion species. The third step is the nucleophilic attack of the carbanion species toward the acetaldehyde adsorbed, thus generating acetaldol. After that, this compound is easily dehydrated to 2-butenal, which is hydrolyzed to butyraldehyde and *n*-butanol. Di Cosimo et al. [4] employing mixed oxides comprising of Mg and Al showed that the condensation of acetaldehyde, i.e., the 2° and 3° steps are the slowest ones.

Both of the prepared mixed oxides are able to synthesize C4 compounds despite the fact that only weak and medium strength basic sites are observed on their surface (Fig. 3). Therefore, strong basic sites seem not to be essential for the proton abstraction from acetaldehyde and ethanol as well.

MgO is not able to promote the ethanol condensation at these experimental conditions in spite of its high density of strong basic sites (Fig. 4 and Table 3). MgO only shows a very low activity for ethanol dehydrogenation. Indeed, MgO generates C4 compounds at much higher temperature. The catalytic behavior of MgO can be associated with its low acidity.

The Al_2O_3 sample shows pairs of strong acid sites and weak basic sites ([14] and Table 3). When employing this oxide only ethylene is observed. One can suggest that as Al_2O_3 does not show medium or strong basic sites it is not able to synthesize C4 compounds as well (Fig. 4).

When the $\text{MgO} + \text{Al}_2\text{O}_3$ physical mixture is used, and even though it shows strong and medium basic and also acid sites, it generates almost only ethylene (Fig. 4). This result shows that the strong or the medium basic sites should be adjacent to the acid sites in order to synthesize C4 compounds.

It can be suggested that the synthesis of C4 compounds is associated with pairs of acid and medium basic sites, which occurs on the mixed oxide surface. The mixed oxides do synthesize C4 compounds due to the insertion of Al in MgO or Mg in $\gamma\text{-Al}_2\text{O}_3$, which generates these adjacent acid and medium basic sites.

The MgAl 1:1 catalyst shows higher density of medium basic sites than MgAl 3:1, which can be seen in Table 3. Considering that the amount of acid sites is enough to carry out the reaction, the differences observed for the ethanol conversion and selectivity to C4 compounds when comparing the catalytic performance of these two samples might be associated with the density of the medium basic sites.

Some authors suggest [4,5,11] that, for the aldol condensation (2° and 3° steps described above), a specific superficial atomic arrangement comprising an ensemble of one basic site adjacent to two acids sites is needed. Moreover, this arrangement must also accommodate the intermediate species of this reaction. According to these authors this system should be able to adsorb two molecules

of acetaldehyde, remove a proton from one of these aldehydes, and promote the nucleophilic attack.

The NMR and XRD results show that MgAl 3:1 and MgAl 1:1 display a completely different structure. Thus, it can be suggested that the superficial atomic arrangement of these catalysts is not similar. However, Fig. 4 depicts that the two mixed oxides show a similar catalytic performance when comparing conversion and selectivity toward the C4 compounds (*n*-butanol and butyraldehyde). Thus, it might be inferred that there is not a strong correlation between the catalytic behavior and the superficial atomic arrangement. Therefore, it may be suggested that the nucleophilic attack mentioned above might occur by an Eley–Rideal mechanism [18]. In this case, the carbanion species generated by the proton abstraction from acetaldehyde desorb and react with another acetaldehyde adsorbed on the acid sites. This mechanism suggests that pairs of acid and basic sites are relevant for this reaction due to the generation of the carbanion species. As the carbanion species do not remain adsorbed, a specific superficial arrangement seems not to be essential.

On the other hand, the synthesized catalysts do show a very distinct selectivity toward butyraldehyde and *n*-butanol (Fig. 4). It can also be suggested that the hydrogenation processes, i.e., the generation of butyraldehyde from 2-butenal and *n*-butanol from butyraldehyde do change with the superficial atom arrangement and/or the Mg/Al ratio.

4. Conclusion

The Mg and Al mixed oxides produced by the thermal decomposition of hydrotalcites are able to promote the synthesis of *n*-butanol from ethanol in a one-step process. It was verified that adjacent acid and medium strength basic sites are needed in order to generate the intermediate compounds of this reaction system. These pairs of acid and basic sites are obtained by the insertion of Al in the MgO lattice or in consequence of the presence of Mg in the γ -Al₂O₃ lattice. Strong basic sites and a specific superficial atomic arrangement seem not to be essential for the C4 synthesis from

ethanol. Mixed oxides with high Mg concentration are the most selective to *n*-butanol.

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References

- [1] M. Schulz, S. Clark, J. ASTM Int. 8 (2011) 1–19.
- [2] C. Yang, Z. Meng, J. Catal. 142 (1993) 37–44.
- [3] R.E. Kirk, D.F. Othmer, Encyclopedia of Chemical Technology, vol. 4, third ed., J.W. Sons, New York, 1978, pp. 338–345.
- [4] J.I. Di Cosimo, C.R. Apesteguía, M.J.L. Ginés, E. Iglesia, J. Catal. 190 (2000) 261–275.
- [5] T. Tsuchida, J. Kubota, T. Yoshioka, S. Sakuma, T. Takeguchi, W. Ueda, J. Catal. 259 (2008) 183–189.
- [6] K. Kourtakos, M.B. D'Amore, L.E. Manzer, U.S. Pat. 0054705 A1, 2009, 1–4.
- [7] S. Ordóñez, E. Díaz, M. León, L. Faba, Catal. Today 167 (2011) 71–76.
- [8] M.R. Othman, Z. Helwani, Martunus, W.J.N. Fernando, Appl. Organomet. Chem. 23 (2009) 335–346.
- [9] M. León, E. Díaz, S. Ordóñez, Catal. Today 164 (2011) 436–442.
- [10] I.-C. Marcu, D. Tichit, F. Fajula, N. Tanchoux, Catal. Today 147 (2009) 231–238.
- [11] V.V. Ordonsky, V.L. Sushkevich, I.I. Ivanova, J. Mol. Catal. A: Chem. 333 (2010) 85–93.
- [12] W.G. Moffatt, G.W. Pearsall, J. Wulff, Ciência dos Materiais, vol. 1, first ed., L.T. Científicos, Rio de Janeiro, 1972, pp. 45–75.
- [13] V.K. Díez, C.R. Apesteguía, J.I. Di Cosimo, J. Catal. 215 (2003) 220–233.
- [14] P.C. Zonetti, J. Celnik, S. Letichevsky, A.B. Gaspar, L.G. Appel, J. Mol. Catal. A: Chem. 334 (2011) 29–34.
- [15] A.S. Ndou, N. Plint, N.J. Coville, Appl. Catal. A: Gen. 251 (2003) 337–345.
- [16] T. Tsuchida, S. Sakuma, T. Takeguchi, W. Ueda, Ind. Eng. Chem. Res. 45 (2006) 8634–8642.
- [17] J.I. Di Cosimo, V.K. Díez, M. Xu, E. Iglesia, C.R. Apesteguía, J. Catal. 178 (1998) 499–510.
- [18] K. Inui, T. Kurabayashi, S. Sato, N. Ichikawa, J. Mol. Catal. A: Chem. 216 (2004) 147–156.